

1 The theta term and Axion Electrodynamics

In relativistic notation, the Maxwell action for electromagnetism takes a wonderfully compact form:

$$S_{Max} = \int d^4x -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} = \int d^4x \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) \quad (1)$$

Here:

$$\left\{ \begin{array}{l} F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \\ = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix} \\ E_i = F_{0i} \\ F_{ij} = -\epsilon_{ijk}B_k \end{array} \right. \quad (2)$$

One reason the Maxwell action is so simple is that there is very little else we can write down that is both gauge-invariant and Lorentz-invariant. There is, however, one additional term that we can add to the Maxwell action that is also gauge-invariant and Lorentz-invariant; we call it the theta term:

$$S_\theta = \frac{\theta e^2}{4\pi^2} \int d^4x \frac{1}{4} \star F^{\mu\nu} F_{\mu\nu} = -\frac{\theta e^2}{4\pi^2} \int d^4x \mathbf{E} \cdot \mathbf{B} \quad (3)$$

where $\star F^{\mu\nu}$ is the dual tensor:

$$\begin{aligned} \star F^{\mu\nu} &= \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \\ &= \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z & -E_y \\ B_y & -E_z & 0 & E_x \\ B_z & E_y & -E_x & 0 \end{pmatrix} \end{aligned} \quad (4)$$

with θ a constant parameter.

However, it is straightforward to check that the theta term can be written as a total derivative:

$$\begin{aligned} \int d^4x \frac{1}{4} \star F^{\mu\nu} F_{\mu\nu} &= \frac{1}{2} \int d^4x \cdot \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} F_{\mu\nu} \\ &= \frac{1}{8} \int d^4x \cdot \epsilon^{\mu\nu\rho\sigma} (\partial_\rho A_\sigma - \partial_\sigma A_\rho) (\partial_\mu A_\nu - \partial_\nu A_\mu) \\ &= \frac{1}{2} \int d^4x \cdot \epsilon^{\mu\nu\rho\sigma} (\partial_\mu A_\nu \partial_\rho A_\sigma) \end{aligned} \quad (5)$$

Now let us prove a useful identity:

$$\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma = 0 \quad (6)$$

because:

$$\begin{aligned}
 \epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma &= \epsilon^{\rho\nu\mu\sigma} A_\nu \partial_\rho \partial_\mu A_\sigma \\
 &= \epsilon^{\rho\nu\mu\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma \\
 &= -\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma
 \end{aligned} \tag{7}$$

Therefore, we obtain:

$$2\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma = 0 \Rightarrow \epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma = 0 \tag{8}$$

Thus, the quadratic part of the theta term becomes:

$$\begin{aligned}
 \int d^4x \frac{1}{4} \star F^{\mu\nu} F_{\mu\nu} &= \frac{1}{2} \int d^4x \epsilon^{\mu\nu\rho\sigma} (\partial_\mu A_\nu \partial_\rho A_\sigma + A_\nu \partial_\mu \partial_\rho A_\sigma) \\
 &= \frac{1}{2} \int d^4x \epsilon^{\mu\nu\rho\sigma} \partial_\mu (A_\nu \partial_\rho A_\sigma)
 \end{aligned} \tag{9}$$

Therefore, the theta term can be written as:

$$S_\theta = \frac{\theta e^2}{8\pi^2} \int d^4x \partial_\mu (\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\rho A_\sigma) \tag{10}$$

We therefore say that the theta term is topological: it depends only on boundary data. However, when deriving the equations of motion via the principle of least action, the fields are held fixed at the boundary (at infinity). Therefore, the theta term does not modify the classical equations of motion and, at first sight, seems to have little physical effect.

But in situations involving a subtle interplay between quantum mechanics and topology, the theta term leads to a number of interesting physical phenomena.

We can also consider situations in which θ affects the dynamics classically. This occurs when θ is not constant, but instead varies in space and time.

$$\theta = \theta(\mathbf{x}, t) \tag{11}$$

We can then write down the action for axion electrodynamics:

$$S = \int d^4x \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{e^2}{16\pi^2} \theta(\mathbf{x}, t) \star F^{\mu\nu} F_{\mu\nu} \right) \tag{12}$$

We use the Euler–Lagrange equations:

$$\frac{\partial \mathcal{L}}{\partial A_\mu} - \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} \right) = 0 \tag{13}$$

to obtain the equations of motion for axion electrodynamics.

Since $\mathcal{L}$ does not depend explicitly on A_μ , the equations of motion reduce to:

$$\frac{\partial \mathcal{L}}{\partial A_\mu} = 0 \Rightarrow \partial_\nu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} \right] = 0 \quad (14)$$

and the Lagrangian density for axion electrodynamics is ($\alpha = \frac{e^2}{4\pi}$):

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{Max} + \mathcal{L}_\theta \\ &= -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{\alpha}{4\pi} \theta(\mathbf{x}, t) \star F^{\mu\nu} F_{\mu\nu} \end{aligned} \quad (15)$$

First, let us consider $\mathcal{L}_\theta$:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} &= \frac{\partial}{\partial (\partial_\nu A_\mu)} \cdot \frac{\alpha}{4\pi} \theta(\mathbf{x}, t) \cdot \frac{1}{2} \epsilon^{\alpha\beta\rho\sigma} F_{\alpha\beta} F_{\rho\sigma} \\ &= \frac{\alpha}{8\pi} \left\{ \left[\frac{\partial}{\partial (\partial_\nu A_\mu)} \theta(\mathbf{x}, t) \right] \cdot \epsilon^{\alpha\beta\rho\sigma} F_{\alpha\beta} F_{\rho\sigma} + \theta(\mathbf{x}, t) \left[\frac{\partial \epsilon^{\alpha\beta\rho\sigma} F_{\alpha\beta} F_{\rho\sigma}}{\partial (\partial_\nu A_\mu)} \right] \right\} \end{aligned} \quad (16)$$

Let us first consider the first term:

$$\epsilon^{\alpha\beta\rho\sigma} F_{\alpha\beta} F_{\rho\sigma} \frac{\partial}{\partial (\partial_\nu A_\mu)} \theta(\mathbf{x}, t) = 0 \quad (17)$$

Next, consider the second term:

$$\begin{aligned} \theta(\mathbf{x}, t) \cdot \frac{\partial}{\partial (\partial_\nu A_\mu)} \cdot \epsilon^{\alpha\beta\rho\sigma} F_{\alpha\beta} F_{\rho\sigma} &= 4\theta(\mathbf{x}, t) \left(\frac{\partial}{\partial (\partial_\nu A_\mu)} \partial_\alpha A_\beta \partial_\rho A_\sigma \right) \epsilon^{\alpha\beta\rho\sigma} \\ &= 4\theta(\mathbf{x}, t) (\delta_\alpha^\nu \delta_\beta^\mu \partial_\rho A_\sigma + \delta_\rho^\nu \delta_\sigma^\mu \partial_\alpha A_\beta) \cdot \epsilon^{\alpha\beta\rho\sigma} \\ &= 4\theta(\mathbf{x}, t) \cdot \epsilon^{\nu\mu\rho\sigma} (\partial_\rho A_\sigma - \partial_\sigma A_\rho) \\ &= 8\theta(\mathbf{x}, t) \star F^{\nu\mu} \end{aligned} \quad (18)$$

Next, we have:

$$\partial_\nu [\theta(\mathbf{x}, t) \star F^{\nu\mu}] = \star F^{\nu\mu} \partial_\nu \theta(\mathbf{x}, t) + \theta(\mathbf{x}, t) \cdot \partial_\nu \star F^{\nu\mu} \quad (19)$$

Using the Bianchi identity:

$$\partial_\mu \star F^{\mu\nu} = 0 \quad (20)$$

so we obtain:

$$\partial_\nu \theta(\mathbf{x}, t) \star F^{\nu\mu} = \star F^{\nu\mu} \partial_\nu \theta(\mathbf{x}, t) \quad (21)$$

Finally, we can get:

$$\partial_\nu \left[\frac{\partial \mathcal{L}_\theta}{\partial (\partial_\nu A_\mu)} \right] = \frac{\alpha}{\pi} \star F^{\nu\mu} \cdot \partial_\nu \theta(\mathbf{x}, t) \quad (22)$$

For the Maxwell term, we have:

$$\partial_\nu \left[\frac{\partial \mathcal{L}_{Max}}{\partial (\partial_\nu A_\mu)} \right] = -\partial_\nu F^{\nu\mu} \quad (23)$$

Therefore, the equations of motion for axion electrodynamics are:

$$\begin{cases} \partial_\nu F^{\nu\mu} = \frac{\alpha}{\pi} \star F^{\nu\mu} \cdot \partial_\nu \theta(x, t) \\ \partial_\mu \star F^{\mu\nu} = 0 \end{cases} \quad (24)$$

In terms of the electric and magnetic fields, the modified Maxwell equations (axion electrodynamics) are:

$$\begin{cases} \nabla \cdot \mathbf{E} = \frac{\alpha}{\pi} (\nabla \theta) \cdot \mathbf{B} \\ \nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t} - \frac{\alpha}{\pi} (\dot{\theta} \mathbf{B} + \nabla \theta \times \mathbf{E}) \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \end{cases} \quad (25)$$

The first equation tells us that in regions of space where θ varies, a magnetic field $\mathbf{B}$ acts like an electric charge density. The second equation tells us that the combination $(\dot{\theta} \mathbf{B} + \nabla \theta \times \mathbf{E})$ acts like a current density.

1.1 The periodicity of theta and its values under parity or time-reversal symmetry

In quantum theory, θ is a periodic variable: it lies in the range:

$$\theta \in [0, 2\pi) \quad (26)$$

After imposing appropriate boundary conditions, S_θ can only take values of the form:

$$S_\theta = \theta N \quad \text{with } N \in \mathbb{Z} \quad (27)$$

This means that the theta angle contributes to the partition function as:

$$e^{iS_\theta} = e^{iN\theta} \quad (28)$$

To show that S_θ must take this form, we consider a compact Euclidean spacetime, which we take to be T^4 , with each circle of the torus having radius R .

We consider the simple case:

$$\begin{cases} \mathbf{E} = (0, 0, E_z), \quad E_z = F_{03} = \partial_0 A_3 - \partial_3 A_0 \\ \mathbf{B} = (0, 0, B_z), \quad B_z = F_{21} = \partial_2 A_1 - \partial_1 A_2 \end{cases} \quad (29)$$

The relevant integral in S_θ is:

$$I = \int_{T^4} d^4x E_z \cdot B_z = \int_{T^4} dx^0 dx^3 \cdot E_z \cdot \int_{T^4} dx^1 dx^2 \cdot B_z \quad (30)$$

The gauge field A_μ must be well defined on the underlying torus, which will put restrictions on the allowed values of E_z and B_z . Thus, the integral cannot take an arbitrary value.

Now let us consider the restrictions on A_μ required for it to be well defined on the torus. When a spatial direction, say x^1 , is periodic with radius R , the physical state at the initial point ($x^1 = 0$) must be identical to that at the final point:

$$\text{state}[A_1(0)] = \text{state}[A_1(2\pi R)] \quad (31)$$

Since $B_z = \partial_2 A_1$ and (taking $A_2 = 0$) A_2 is constant, we have $A_1 = B_z x^2 + C$. The simplest choice $A_1 = 0$ is trivial. Therefore, we should find a gauge transformation that connects $A_1(0)$ and $A_1(2\pi R)$, so that two different values of A_1 correspond to the same physical state.

$$A_1(2\pi R) = A_1(0) + G \quad (32)$$

Let us find the gauge transformation of interest:

$$A_1 \rightarrow A_1 + \partial_1 \chi \quad (33)$$

We do not require $\chi(x)$ to be single valued. Instead, we require only that $e^{ie\chi}$ be single-valued, since this is what acts on the wavefunction. Therefore, χ can take the simple form:

$$\chi = \frac{nx^1}{eR} \quad \text{with } n \in \mathbb{Z} \quad (34)$$

Because at the same point ($2\pi NR + X$), $e^{ie\chi}$ represents the same state.

These are sometimes called large gauge transformations, a name which reflects the fact that they cannot be continuously deformed to the identity. Under such a gauge transformation, we have:

$$A_1 \rightarrow A_1 + \partial_1 \chi = A_1 + \frac{n}{eR} \quad (35)$$

The two different values of A_1 correspond to the same physical state. From this transformation, we can determine the expression for A_1 :

$$\begin{aligned} A_1(0) &= A_1(2\pi R) + \frac{n}{eR} \\ \Rightarrow A_1 &= \frac{nx^2}{2\pi eR^2} \end{aligned} \quad (36)$$

We can then write down the expression for B_z :

$$B_z = \partial_2 A_1 = \frac{n}{2\pi eR^2} \quad (37)$$

Therefore, the integral of the magnetic field is:

$$\int_{T^2} dx^1 dx^2 B_z = \frac{2\pi n}{e} \quad \text{with } n \in \mathbb{Z} \quad (38)$$

We can also apply exactly the same argument to the electric field.

$$E_z = \partial_0 A_3 \quad (A_0 = 0) \quad (39)$$

So the integral of electric field is:

$$\int_{T^2} dx^0 dx^3 E_z = \frac{2\pi n'}{e} \quad \text{with } n' \in \mathbb{Z} \quad (40)$$

Armed with two integrals above, the S_θ becomes:

$$\begin{aligned} S_\theta &= -\frac{\theta e^2}{4\pi^2} \int_{T^4} d^4x \mathbf{E} \cdot \mathbf{B} \\ &= -\frac{\theta e^2}{4\pi^2} \frac{4\pi^2}{e^2} \cdot nn' \quad (\text{let } N = -nn') \\ &= N\theta \quad \text{with } N \in \mathbb{Z} \end{aligned} \quad (41)$$

It is straightforward to generalise this to non-constant $\mathbf{E}$ and $\mathbf{B}$ fields:

$$\begin{aligned} \int_{T^2} dx^1 dx^2 B_z &= \int_{T^2} dx^1 dx^2 \cdot \partial_2 A_1 \\ &= 2\pi R [A_1(2\pi R) - A_1(0)] \\ &= \frac{2\pi n}{e} \end{aligned} \quad (42)$$

Therefore, θ is a periodic variable in quantum theory.

Although the theta term is gauge-invariant and Lorentz-invariant, it is not invariant under certain discrete symmetries and therefore does not preserve the same symmetries as the Maxwell term. These discrete symmetries are parity $\mathcal{P}$ and time reversal $\mathcal{T}$.

$$\begin{cases} \mathcal{P} : \mathbf{x} \rightarrow -\mathbf{x} \\ \mathcal{T} : t \rightarrow -t \end{cases} \quad (43)$$

If we act on both $\mathbf{E}$ and $\mathbf{B}$, we have

$$\begin{cases} \mathcal{P} : \mathbf{E}(\mathbf{x}, t) \rightarrow -\mathbf{E}(-\mathbf{x}, t) \text{ and } \mathcal{P} : \mathbf{B}(\mathbf{x}, t) \rightarrow \mathbf{B}(-\mathbf{x}, t) \\ \mathcal{T} : \mathbf{E}(\mathbf{x}, t) \rightarrow \mathbf{E}(\mathbf{x}, -t) \text{ and } \mathcal{T} : \mathbf{B}(\mathbf{x}, t) \rightarrow -\mathbf{B}(\mathbf{x}, -t) \end{cases} \quad (44)$$

So $\mathbf{E}$ is odd under parity and even under time reversal. $\mathbf{B}$ is even under parity and odd under time reversal.

This means that the theta term $S_\theta \propto \int d^4x \mathbf{E} \cdot \mathbf{B}$ breaks both parity and time-reversal invariance: one finds $S_\theta \rightarrow -S_\theta$ under $\mathcal{P}$ and $\mathcal{T}$. However, because of the periodicity of θ , there are two exceptions:

$$S_\theta = -S_\theta + 2\pi N \quad (45)$$

To make this equation hold for any integer value of N , θ must be equal to 0 or π ($\theta \in [0, 2\pi)$).

Therefore, when $\theta = 0$ or $\theta = \pi$, the theory is invariant under $\mathcal{P}$ and $\mathcal{T}$ symmetries.

In subsequent sections, we will see that a topological insulator with a $\mathbb{Z}_2$ classification has $\theta = \pi$ because such materials are defined to be time-reversal invariant. Most materials have $\theta = 0$. However, some materials have a band structure that is twisted in a particular way, which results in $\theta = \pi$.

1.2 Continuity Conditions

Let us consider an equivalent way to describe the physics above. We introduce the electric displacement:

$$\mathbf{D} = \mathbf{E} - \frac{\alpha\theta}{\pi} \cdot \mathbf{B} \quad (46)$$

Because the divergence of $\mathbf{B}$ is equal to zero, we have:

$$\nabla \cdot \mathbf{D} = \nabla \cdot \mathbf{E} - \frac{\alpha}{\pi} (\nabla \cdot \theta) \mathbf{B} = 0. \quad (47)$$

Comparing with the expression for $\mathbf{D}$, we see that a magnetic field $\mathbf{B}$ acts like a polarisation in a topological insulator. When θ varies, we obtain a spatially varying polarisation, which results in bound charge. Similarly, we define the magnetising field:

$$\mathbf{H} = \mathbf{B} + \frac{\alpha}{\pi} \theta \cdot \mathbf{E} \quad (48)$$

Using the relationship between $\mathbf{B}$ and $\mathbf{E}$:

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0 \quad (49)$$

So we obtain:

$$\begin{aligned} \nabla \times \mathbf{H} &= \nabla \times \mathbf{B} + \frac{\alpha}{\pi} [(\nabla \theta) \times \mathbf{E} + \theta (\nabla \times \mathbf{E})] \\ &= \frac{\partial \mathbf{D}}{\partial t} + \frac{\alpha}{\pi} \theta \cdot \left(\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} \right) \\ &= \frac{\partial \mathbf{D}}{\partial t} \end{aligned} \quad (50)$$

In a topological insulator, $\mathbf{E}$ acts like a magnetisation. When θ varies, we obtain a spatially varying magnetisation, which results in bound currents.

With these definitions, the equations of axion electrodynamics take the usual form of Maxwell equations in matter:

$$\begin{cases} \nabla \cdot \mathbf{D} = 0 \\ \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = 0 \\ \nabla \cdot \mathbf{B} = 0 \\ \frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0 \end{cases} \quad (51)$$

This means that if we introduce the surface normal vector $\hat{\mathbf{n}}$, we have the boundary conditions:

$$\begin{cases} \hat{\mathbf{n}} \cdot \Delta \mathbf{D} = 0 \\ \hat{\mathbf{n}} \times \Delta \mathbf{E} = 0 \\ \hat{\mathbf{n}} \cdot \Delta \mathbf{B} = 0 \\ \hat{\mathbf{n}} \times \Delta \mathbf{H} = 0 \end{cases} \quad (52)$$

1.3 The Witten effect

The effect of the θ term is to endow a magnetic monopole with an electric charge; this is known as the Witten effect.

Let us take a magnetic monopole with magnetic charge g and place it inside a vacuum ($\theta = 0$). We then surround it with a medium that has $\theta = \Theta \neq 0$.

When the magnetic field crosses the interface where θ changes, it will induce an electric charge:

$$\begin{aligned} \int \nabla \cdot \mathbf{E} dv &= \int \frac{\alpha}{\pi} (\nabla \cdot \theta) \cdot \mathbf{B} dv. \\ \Rightarrow Q &= \frac{\alpha}{\pi} \Theta g \end{aligned} \quad (53)$$

This result is independent of the size of the interior region where θ is zero. We could therefore shrink this region down until it is infinitesimally small, and we would still find that the monopole carries charge q . In fact, when θ is not equal to zero, a monopole is a dyon that carries both magnetic and electric charge.

When the monopole carries the minimum allowed magnetic charge, its electric charge is given by:

$$g = \frac{2\pi}{e} \Rightarrow q = \frac{e\theta}{2\pi} \quad (54)$$

In particular, if we place a magnetic monopole inside a topological insulator, it turns into a dyon that carries half the charge of an electron.

We can also consider placing a monopole in a medium whose θ will gradually increase from

zero. At the end of this process, the final electric field will be:

$$\begin{aligned} \int_0^{t_0} \frac{\partial \mathbf{E}}{\partial t} dt &= \frac{\alpha}{\pi} \int_0^{t_0} \dot{\theta} \mathbf{B} \\ \Rightarrow \mathbf{E} &= \frac{\alpha}{\pi} \Theta \mathbf{B} \quad [\text{which } \theta(t_0) = \Theta] \\ \Rightarrow q &= \frac{\alpha}{\pi} \Theta g \end{aligned} \quad (55)$$

1.4 A Mirage Monopole

Let us consider an electric charge q placed in the vacuum at the point $\mathbf{x} = (0, 0, d)$. We then fill the region where z is less than zero with a topological insulator. One finds that the effect produced by the surface charge is equivalent to placing mirror dyons at the points $\mathbf{x} = (0, 0, d)$ and $\mathbf{x}' = (0, 0, -d)$.

We assume that the mirror dyon at $\mathbf{x} = (0, 0, d)$ has electric and magnetic charges (q_1, g_1) and that the mirror dyon at $\mathbf{x}' = (0, 0, -d)$ has electric and magnetic charges (q_2, g_2) .

Next, we write:

$$\begin{cases} \mathbf{E} = -\nabla \phi & (\text{because } \nabla \times \mathbf{E} = 0) \\ \mathbf{B} = -\nabla \Omega & (\text{because } \nabla \times \mathbf{B} = 0) \end{cases} \quad (56)$$

The electric potential in the two regions is:

$$\begin{cases} \phi_+ = \frac{1}{4\pi} \left[\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} + \frac{q_2}{\sqrt{x^2 + y^2 + (z+d)^2}} \right], & z > 0 \\ \phi_- = \frac{1}{4\pi} \frac{q + q_1}{\sqrt{x^2 + y^2 + (z-d)^2}}, & z < 0 \end{cases} \quad (57)$$

By symmetry, we only need to consider the electric and magnetic fields in the yz plane at $z = 0$.

$$\begin{cases} \left. \begin{aligned} E_y^+ &= -\partial_y \phi_+ = \frac{q + q_2}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ E_z^+ &= -\partial_z \phi_+ = \frac{q_2 - q}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{aligned} \right\} z = 0^+ \\ \left. \begin{aligned} E_y^- &= -\partial_y \phi_- = \frac{q + q_1}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ E_z^- &= -\partial_z \phi_- = \frac{-(q + q_1)}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{aligned} \right\} z = 0^- \end{cases} \quad (58)$$

Using the boundary conditions:

$$\hat{\mathbf{n}} \times \mathbf{E} = 0 \quad (59)$$

So we obtain:

$$E_y^+ = E_y^- \Rightarrow q_1 = q_2 = q' \quad (60)$$

We can perform the same procedure for the magnetic field. The magnetic scalar potential is:

$$\begin{cases} \Omega_+ = \frac{1}{4\pi} \cdot \frac{g_2}{\sqrt{x^2 + y^2 + (z+d)^2}}, & z > 0 \\ \Omega_- = \frac{1}{4\pi} \cdot \frac{g_1}{\sqrt{x^2 + y^2 + (z-d)^2}}, & z < 0 \end{cases} \quad (61)$$

Thus, the magnetic field in the yz plane at $z = 0$ is:

$$\begin{cases} \left. \begin{aligned} B_y^+ &= -\partial_y \Omega_+ = \frac{g_2}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ B_z^+ &= -\partial_z \Omega_+ = \frac{g_2}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{aligned} \right\} z = 0^+ \\ \left. \begin{aligned} B_y^- &= -\partial_y \Omega_- = \frac{g_1}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ B_z^- &= -\partial_z \Omega_- = \frac{-g_1}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{aligned} \right\} z = 0^- \end{cases} \quad (62)$$

Using the boundary conditions:

$$\hat{n} \cdot \mathbf{B} = 0 \Rightarrow B_z^- = B_z^+ \Rightarrow g_2 = -g_1 = g \quad (63)$$

From the expression above, we obtain the electric displacement:

$$\begin{cases} D_z^+ = E_z^+ = \frac{q' - q}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ D_z^- = E_z^- - \frac{\alpha}{\pi} \theta B_z^- = \frac{-(q + q') + \alpha g}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{cases} \quad (64)$$

Using the boundary conditions:

$$\hat{n} \cdot \mathbf{D} = 0 \Rightarrow D_z^+ = D_z^- \Rightarrow g = \frac{-2q'}{\alpha} \quad (65)$$

The magnetising field:

$$\begin{cases} H_y^+ = B_y^+ = \frac{g}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ H_y^- = B_y^- + \frac{\alpha}{\pi} \theta E_y^- = \frac{-g + \alpha(q + q')}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{cases} \quad (66)$$

Using the boundary condition:

$$\nabla \times \mathbf{H} = 0 \Rightarrow H_y^+ = H_y^- \Rightarrow g = \frac{\alpha(q + q')}{2} \quad (67)$$

Finally, we obtain the electric and magnetic charges carried by these dyons:

$$\begin{cases} q' = \frac{-\alpha^2}{4 + \alpha^2} \cdot q \\ g = \frac{2\alpha}{4 + \alpha^2} \cdot q \end{cases} \quad (68)$$

2 3D Topological Insulator

Let us consider the low-energy continuum model for prototypical 3D topological insulators such as Bi_2Se_3 . The bulk electronic structure of Bi_2Se_3 near the Fermi level is described by two p orbitals, $P1_z^+$ and $P2_z^-$, where $\pm$ denote parity. Defining the basis $\{|P1_z^+, \uparrow\rangle, |P1_z^+, \downarrow\rangle, |P2_z^-, \uparrow\rangle, |P2_z^-, \downarrow\rangle\}$ and retaining the wave vector $\mathbf{k}$ up to quadratic order, the low-energy effective Hamiltonian around the Γ point is given by:

$$H_{\text{eff}}(\mathbf{k}) = \begin{bmatrix} M(\mathbf{k}) & 0 & A_1 k_z & A_2 k_- \\ 0 & M(\mathbf{k}) & A_2 k_+ & -A_1 k_z \\ A_1 k_z & A_2 k_- & -M(\mathbf{k}) & 0 \\ A_2 k_+ & -A_1 k_z & 0 & -M(\mathbf{k}) \end{bmatrix} \quad (69)$$

$$= A_2 k_x \alpha_1 + A_2 k_y \alpha_2 + A_1 k_z \alpha_3 + M(\mathbf{k}) \alpha_4$$

where $k_{\pm} = k_x \pm i k_y$ and $M(\mathbf{k}) = m_0 - B_1 k_z^2 - B_2(k_x^2 + k_y^2)$. The 4×4 matrices α_{μ} are given in the Dirac representation:

$$\alpha_j = \begin{bmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{bmatrix}, \quad \alpha_4 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (70)$$

Next, let us consider the lattice model corresponding to the Hamiltonian above, which allows us to compute the $\mathbb{Z}_2$ invariant and identify whether a phase is topologically nontrivial or trivial.

The simplest 3D lattice is the cubic lattice. We make the following substitutions (ignoring the lattice constant a):

$$\begin{cases} k_i \rightarrow \sin k_i \\ k_i^2 \rightarrow 2(1 - \cos k_i) \end{cases} \quad (71)$$

and simplify the coefficients to obtain an isotropic lattice Hamiltonian:

$$H_{\text{eff}}(\mathbf{k}) = A(\alpha_1 \sin k_x + \alpha_2 \sin k_y + \alpha_3 \sin k_z) + [m_0 + r \sum (1 - \cos k_i)] \alpha_4 \quad (72)$$

where we define $A = A_1 = A_2$ and $r = -2B_1 = -2B_2$. We then use the Fu–Kane method to calculate the $\mathbb{Z}_2$ topological invariant for this system:

$$(-1)^{\nu} = \prod_{i=1}^{N_{\text{TRIM}}} \text{sgn}[M(\Lambda_i)] \quad (73)$$

In this 3D cubic lattice, there are eight TRIMs (time-reversal-invariant momenta, which are invariant under $k_i \rightarrow -k_i$):

$$\begin{cases} \Gamma : (0, 0, 0) \\ X : (\pi, 0, 0) / (0, \pi, 0) / (0, 0, \pi) \\ M : (\pi, \pi, 0) / (\pi, 0, \pi) / (0, \pi, \pi) \\ R : (\pi, \pi, \pi) \end{cases} \quad (74)$$

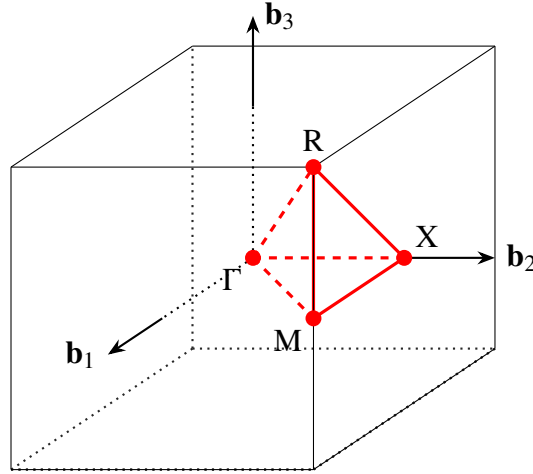


Figure 1: Highly symmetric points in the Brillouin zone of a cubic lattice

The mass term is:

$$M(\mathbf{k}) = m_0 + r \sum_{t=x,y,z} (1 - \cos k_t) \quad (75)$$

so we can compute the mass term for each group of points:

$$\begin{cases} \Gamma : M = m_0 \\ X : M = m_0 + 2r \\ M : M = m_0 + 4r \\ R : M = m_0 + 6r \end{cases} \quad (76)$$

Now we substitute these mass terms into the $\mathbb{Z}_2$ topological-invariant formula:

$$\begin{aligned} (-1)^\nu &= \text{sgn}(m_0) \times \text{sgn}(m_0 + 2r)^3 \times \text{sgn}(m_0 + 4r)^3 \times \text{sgn}(m_0 + 6r) \\ &= \text{sgn}(m_0) \times \text{sgn}(m_0 + 2r) \times \text{sgn}(m_0 + 4r) \times \text{sgn}(m_0 + 6r) \\ &= \begin{cases} -1 & (0 > \frac{m_0}{r} > -2, -4 > \frac{m_0}{r} > -6) \\ 1 & (\frac{m_0}{r} > 0, -2 > \frac{m_0}{r} > -4, -6 > \frac{m_0}{r}) \end{cases} \end{aligned} \quad (77)$$

Let us return to the isotropic continuum Hamiltonian to obtain the θ term. The Hamiltonian around the Γ point can be simplified by ignoring terms that are second order in k_i :

$$H_{\text{TI}}(\mathbf{k}) = A\mathbf{k} \cdot \boldsymbol{\alpha} + m_0\alpha_4 \quad (78)$$

Because this Hamiltonian describes a topological insulator, m_0 is less than zero ($(-1)^\nu = \text{sgn}(m_0)$). Apart from the negative mass m_0 , this is the usual Dirac Hamiltonian ($A\mathbf{k} \rightarrow c\mathbf{p}$). In the presence of an external electromagnetic vector potential $\mathbf{A}$, minimal coupling results in:

$$\mathbf{p} \rightarrow \mathbf{p} + e\mathbf{A} \quad (79)$$

where $e > 0$ is the magnitude of the electron charge. At the same time, in the presence of an external electromagnetic scalar potential A_0 , the energy density is modified as:

$$\psi^\dagger H_0 \psi \rightarrow \psi^\dagger (H_0 - eA_0) \psi \quad (80)$$

Therefore, the action of the system in the presence of an external electromagnetic four-potential $A_\mu = (A_0, -\mathbf{A})$ can be written in the following relativistic form:

$$\begin{aligned} S_{\text{TI}} &= \int dt d^3x \psi^\dagger \{i(\partial_t - ieA_0) - [H_{\text{TI}}(\mathbf{k} + e\mathbf{A})]\} \psi \\ &= \int dt d^3x \psi^\dagger [i(\partial_t - ieA_0) - (-i\nabla + e\mathbf{A}) \cdot \boldsymbol{\alpha} - m_0\alpha_4] \psi \end{aligned} \quad (81)$$

where $\psi^\dagger(x, t)$ is a fermionic field in the basis used to write the Hamiltonian.

We now convert from the basis of $\psi^\dagger$ and the α_j matrices to the relativistic basis of $\bar{\psi}$ and the γ matrices:

$$\left\{ \begin{aligned} \gamma^0 &= \alpha_4 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ \gamma^j &= \alpha_4 \alpha_j = \begin{bmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{bmatrix} \\ \gamma^5 &= i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}, \\ \bar{\psi} &= \psi^\dagger \gamma^0 \end{aligned} \right. \quad (82)$$

To introduce $\bar{\psi}$, we use the identity $(\gamma^0)^2 = \mathbb{I}_{4 \times 4}$, so the Lagrangian density can be rewritten as:

$$\begin{aligned} \mathcal{L}_{\text{TI}} &= \bar{\psi} \gamma^0 [i(\partial_t - ieA_0) + i\nabla \cdot \boldsymbol{\alpha} - e\mathbf{A} \cdot \boldsymbol{\alpha} - m_0\alpha_4] \psi \\ &= \bar{\psi} [\gamma^0(i\partial_t + i\nabla \cdot \boldsymbol{\alpha}) + \gamma^0(eA_0 - e\mathbf{A} \cdot \boldsymbol{\alpha}) - m_0\gamma^0\alpha_4] \psi \\ &= \bar{\psi} [i(\gamma^0\partial_t + \boldsymbol{\gamma} \cdot \boldsymbol{\nabla}) + (eA_0\gamma^0 - e\mathbf{A} \cdot \boldsymbol{\gamma}) - m_0] \psi \\ &= \bar{\psi} [i\gamma^\mu \partial_\mu + e\gamma^\mu A_\mu - m_0] \psi \\ &= \bar{\psi} [i\gamma^\mu (\partial_\mu - ieA_\mu) - m_0] \psi \end{aligned} \quad (83)$$

To facilitate the analysis, we transform from Minkowski spacetime to Euclidean spacetime:

$$S_{\text{TI}}^E = -iS_{\text{TI}}. \quad (84)$$

We rewrite t , A_0 , and γ^j as follows:

$$\left\{ \begin{aligned} t &\rightarrow -i\tau \\ A_0 &\rightarrow iA_0 \\ \gamma^j &\rightarrow i\gamma_j \\ m_0 &= -m_0(\cos \pi + i\gamma_5 \sin \pi) = -m_0 e^{i\pi\gamma_5} \end{aligned} \right. \quad (85)$$

Therefore, the Euclidean-spacetime action of the system is:

$$S_{\text{TI}}^E = \int d\tau d^3x \bar{\psi} [\gamma_\mu (\partial_\mu - ieA_\mu) - m_0 e^{i\pi\gamma_5}] \psi \quad (86)$$

Now we apply Fujikawa's method to the action. First, let us consider an infinitesimal chiral transformation defined by:

$$\begin{cases} \psi \rightarrow \psi' = e^{-\frac{i\pi\phi}{2}\gamma_5} \psi \\ \bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} e^{-\frac{i\pi\phi}{2}\gamma_5} \end{cases} \quad (87)$$

where $\phi \in [0, 1]$. The partition function Z transforms as:

$$\begin{aligned} Z &= \int D[\psi, \bar{\psi}] e^{-S_{\text{TI}}^E[\psi, \bar{\psi}]} \\ \Rightarrow Z' &= \int D[\psi', \bar{\psi}'] e^{-S_{\text{TI}}^E[\psi', \bar{\psi}']} \end{aligned} \quad (88)$$

Let us first consider the first term of $S_{\text{TI}}^{E'}$:

$$K = \int d\tau d^3x \bar{\psi} e^{-i\theta\gamma_5} [\gamma_\mu (\partial_\mu - ieA_\mu)] e^{-i\theta\gamma_5} \psi \quad (89)$$

where $\theta = \frac{\pi d\phi}{2}$. Because the anticommutation relation $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$ holds, we have:

$$e^{-i\theta\gamma_5} \gamma_\mu = \gamma_\mu e^{i\theta\gamma_5} \quad (90)$$

The Lagrangian density of K can be written as:

$$\begin{aligned} \mathcal{L}_K &= \bar{\psi} \gamma_\mu e^{i\theta\gamma_5} (\partial_\mu - ieA_\mu) e^{-i\theta\gamma_5} \psi \\ &= \bar{\psi} \gamma_\mu (\partial_\mu - ieA_\mu) \psi \end{aligned} \quad (91)$$

Next, let us consider the mass term of $S_{\text{TI}}^{E'}$:

$$\begin{aligned} \mathcal{L}_M &= -\bar{\psi} e^{-i\theta\gamma_5} m_0 e^{i\pi\gamma_5} e^{-i\theta\gamma_5} \psi \\ &= -m_0 \bar{\psi} e^{i\pi(1-d\phi)\gamma_5} \psi \end{aligned} \quad (92)$$

Therefore, the action in Euclidean spacetime is transformed as:

$$S_{\text{TI}}^{E'} = \int d\tau d^3x \bar{\psi} [\gamma_\mu (\partial_\mu - ieA_\mu) - m_0 e^{i\pi(1-d\phi)\gamma_5}] \psi \quad (93)$$

We should also consider the transformation of $D[\psi', \bar{\psi}']$.

$$D[\psi', \bar{\psi}'] = JD[\psi, \bar{\psi}] \quad (94)$$

The Jacobian J is written as:

$$\ln J = -i \int d\tau d^3x d\phi \frac{\pi e^2}{16\pi^2} \star F^{\mu\nu} F_{\mu\nu} \quad (95)$$

We integrate with respect to the variable ϕ from 0 to 1. Due to the invariance of the partition function, we finally arrive at the following expression for S_{TI}^E :

$$S_{TI}^E = \int d\tau d^3x \bar{\psi} [\gamma_\mu (\partial_\mu - ieA_\mu) - m_0] \psi + i \int d\tau d^3x \frac{\pi e^2}{16\pi^2} {}^\star F^{\mu\nu} F_{\mu\nu} \quad (96)$$

The first term is the action of a topologically trivial insulator since the mass $-m_0$ is positive. The second term is the θ term, with θ equal to π in imaginary time. Thus, we have demonstrated that for 3D topological insulators, the θ in the θ term is π .

2.1 Topological magnetoelectric effect

We have seen that topological insulators are a fascinating class of materials whose dynamics are characterised by the fact that $\theta = \pi$. We will see that if a topological insulator is placed in an electric or magnetic field, it will exhibit many novel physical effects.

First, we place a topological insulator in a uniform magnetic field directed along the z axis. Because θ undergoes a discontinuity at the surface of the topological insulator, charge is induced on the surface:

$$\int \sigma \cdot ds = \int \nabla \cdot \mathbf{E} dV = \int \frac{\alpha}{\pi} (\nabla \theta) \cdot \mathbf{B} dV. \quad (97)$$

At the upper interface, θ jumps from π to 0, whereas at the lower interface, θ jumps from 0 to π , so:

$$\sigma = \begin{cases} -\alpha B & (\text{upper interface}) \\ \alpha B & (\text{lower interface}) \end{cases} \quad (98)$$

Thus, the magnetic field induces negative charge on the upper interface and positive charge on the lower interface.

Alternatively, we can apply an electric field tangential to the interface by placing a topological insulator in a uniform electric field directed along the y axis. Similarly, because θ undergoes a discontinuity at the surface of the topological insulator, a surface current $\mathbf{K}$ is induced:

$$\int \mathbf{K} \cdot d\mathbf{l} = \int \nabla \times \mathbf{B} ds = - \int \frac{\alpha}{\pi} \nabla \theta \times \mathbf{E} ds \quad (99)$$

For simplicity, we consider only the surface current on the upper interface, so:

$$\begin{aligned} \mathbf{K} &= -\alpha E \hat{e}_x \\ &= -\frac{1}{2} \cdot \frac{e^2}{2\pi} E \hat{e}_x \end{aligned} \quad (100)$$

The creation of a two-dimensional current that lies perpendicular to an applied electric field is called the Hall effect; the coefficient of proportionality is known as the Hall conductivity:

$$\sigma_{xy} = \frac{1}{2} \cdot \frac{e^2}{2\pi} \quad (101)$$

This is usually abbreviated to say that the interface of the topological insulator has Hall conductivity $1/2$. The general phenomenon in which electric fields induce magnetic fields and vice versa goes by the name of the topological magneto-electric effect.

3 Weyl semimetals

Weyl semimetals have nondegenerate, gapless linear dispersions around band-touching points (Weyl nodes). The low-energy effective Hamiltonian around a Weyl node can be written as:

$$H_W(\mathbf{k}) = Qv_F \mathbf{k} \cdot \boldsymbol{\sigma} \quad (102)$$

where $Q = \pm 1$ indicates the chirality, v_F is the Fermi velocity, and σ_i are Pauli matrices. Because the sum of chiralities of Weyl nodes in a system must be zero, the simplest realization of a Weyl semimetal contains two Weyl nodes of opposite chirality. Note that the minimal number of Weyl nodes in a Weyl semimetal with broken inversion symmetry is four, whereas it is two in Weyl semimetals with broken time-reversal symmetry.

Now, let us consider a 4×4 continuum-model Hamiltonian for two-node Weyl semimetals with broken time-reversal symmetry:

$$H_0(\mathbf{k}) = v_F(\tau_z \otimes \mathbf{k} \cdot \boldsymbol{\sigma} + \Delta \tau_x \otimes \mathbb{I} + \mathbb{I} \otimes \mathbf{b} \cdot \boldsymbol{\sigma}) \quad (103)$$

where τ_i and σ_i are Pauli matrices for the Weyl-node and spin degrees of freedom, respectively, and Δ is the mass of 3D Dirac fermions. The term $\mathbf{b} \cdot \boldsymbol{\sigma}$ represents a magnetic interaction, such as exchange coupling between conduction electrons and magnetic impurities, or Zeeman coupling to an external magnetic field. Note that the Hamiltonian with $\mathbf{b} = 0$ describes a topological or normal insulator depending on the sign of Δ (the mass). Without loss of generality, we set $\mathbf{b} = (0, 0, b)$.

Next, let us find the locations of the Weyl nodes. The Hamiltonian can be written as:

$$\begin{aligned} H &= v_F \tau_z (k_x \sigma_x + k_y \sigma_y + k_z \sigma_z) + \Delta \tau_x \otimes \mathbb{I} + b \mathbb{I} \otimes \sigma_z \\ &= \underbrace{v_F \tau_z \otimes (k_x \sigma_x + k_y \sigma_y)}_{H_\perp} + \underbrace{v_F \tau_z \otimes k_z \sigma_z + \Delta \tau_x \otimes \mathbb{I} + b \mathbb{I} \otimes \sigma_z}_{H_\parallel} \end{aligned} \quad (104)$$

Because $H_\perp$ and $H_\parallel$ anticommute, we have:

$$\begin{aligned} H^2 &= H_\perp^2 + H_\parallel^2 + \{H_\perp, H_\parallel\} \\ &= H_\perp^2 + H_\parallel^2 \end{aligned} \quad (105)$$

This leads to the energy spectrum:

$$E^2 = v_F^2 (k_x^2 + k_y^2) + E_\parallel^2 \quad (106)$$

and $H_\parallel$ can be split into two 2×2 blocks:

$$\begin{aligned} h_{eff} &= s(k_z \tau_z + b) + \Delta \tau_x \\ &= s \mathbf{k}_z \cdot \boldsymbol{\tau}_z + \Delta \tau_x + s b \mathbb{I} \end{aligned} \quad (107)$$

where $s = \pm 1$. The energy spectrum of h_{eff} is then:

$$E_{\parallel} = sb \pm \sqrt{k_z^2 + \Delta^2} \quad (108)$$

Therefore, the Weyl-node locations satisfy:

$$\begin{cases} k_x^2 + k_y^2 = 0 \\ sb \pm \sqrt{v_F^2 k_z^2 + \Delta^2} = 0 \end{cases} \quad (109)$$

Thus, the coordinates of the Weyl nodes are:

$$(k_x, k_y, k_z) = (0, 0, \pm \frac{\sqrt{b^2 - \Delta^2}}{v_F}) \quad (110)$$

which requires $b^2 > \Delta^2$.

We now derive the θ term of Weyl semimetals from the microscopic four-band model. To describe a more generic Weyl semimetal, we add the term $\mu_5 \tau_z$ to the Hamiltonian, which generates a chemical-potential difference $2\mu_5$ between the two Weyl nodes. We also set $\Delta = 0$ for simplicity. The Hamiltonian can then be written as:

$$H'_0 = v_F(\tau_z \otimes \boldsymbol{\sigma} \cdot \mathbf{k} + \mathbb{I} \otimes \boldsymbol{\sigma} \cdot \mathbf{b}) - \mu_5 \tau_z \otimes \mathbb{I} \quad (111)$$

We rewrite the Hamiltonian using Gamma matrices.

$$\begin{cases} \gamma^0 = \tau_x \otimes \mathbb{I} \\ \gamma^i = -i\tau_y \otimes \sigma_i \\ \gamma^5 = \tau_z \otimes \mathbb{I} \\ b_\mu = (\mu_5, -\mathbf{b}) \end{cases} \quad (112)$$

and we set $v_F \approx c = 1$, so:

$$H = -\gamma^0 \gamma^i k_i + \gamma^0 \gamma^i \gamma^5 b_i - \mu_5 \gamma^5 \quad (113)$$

The action of the system in the presence of external electric and magnetic fields, with the four-potential $A_\mu = (A_0, -\mathbf{A})$, is given by:

$$\begin{aligned} S &= \int dt d^3x \psi^\dagger [i(\partial_t - ieA_0) - H'_0(\mathbf{k} + e\mathbf{A})] \psi \\ &= \int dt d^3x \bar{\psi} \gamma^0 [i\partial_t + i\gamma^0 \gamma^i \partial_i + eA_0 - e\gamma^0 \gamma^i A_i - \gamma^0 \gamma^i \gamma^5 b_i + \mu_5 \gamma^5] \psi \\ &= \int dt d^3x \bar{\psi} i [\gamma^0 \partial_0 + \gamma^i \partial_i - ie(\gamma^0 A_0 - \gamma^i A_i) - i(\gamma^0 \mu_5 - \gamma^i b_i) \gamma^5] \psi \\ &= \int dt d^3x \bar{\psi} i (\gamma^\mu \partial_\mu - ie\gamma^\mu A_\mu - i\gamma^\mu b_\mu \gamma^5) \psi \\ &= \int dt d^3x \bar{\psi} i \gamma^\mu (\partial_\mu - ieA_\mu - ib_\mu \gamma^5) \psi \end{aligned} \quad (114)$$

where $e > 0$, ψ is a four-component spinor, and $\bar{\psi} = \psi^\dagger \gamma^0$.

Now we apply Fujikawa's method to the action. We perform an infinitesimal chiral transformation repeatedly such that:

$$\begin{cases} \psi \rightarrow \psi' = e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \psi \\ \bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \end{cases} \quad (115)$$

with $\theta(\mathbf{r}, t) = 2x^\mu b_\mu = 2(\mathbf{b} \cdot \mathbf{r} - \mu_5 t)$ and $\phi \in [0, 1]$. The partition function Z is then transformed as:

$$\begin{aligned} Z &= \int D[\psi, \bar{\psi}] e^{iS[\psi, \bar{\psi}]} \\ \Rightarrow Z' &= \int D[\psi', \bar{\psi}'] e^{iS[\psi', \bar{\psi}']} \end{aligned} \quad (116)$$

Because the anticommutation relation $\{\gamma^\mu, \gamma^5\} = 0$ holds, we have:

$$e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \gamma^\mu = \gamma^\mu e^{\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \quad (117)$$

The Lagrangian density of $S[\psi', \bar{\psi}']$ can be written as:

$$\begin{aligned} \mathcal{L} &= \bar{\psi} e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \cdot i\gamma^\mu (\partial_\mu - ieA_\mu - ib_\mu \gamma^5) e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \psi \\ &= \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi + \bar{\psi} i\gamma^\mu (\partial_\mu e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}}) \psi - \bar{\psi} i\gamma^\mu (-ib_\mu \gamma^5) \psi \\ &= \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi - \bar{\psi} i\gamma^\mu (1 - d\phi) ib_\mu \gamma^5 \psi \end{aligned} \quad (118)$$

The Jacobian J of the transformation, defined by $D[\psi', \bar{\psi}'] = JD[\psi, \bar{\psi}]$, is given by:

$$\ln J = i d\phi \int dt d^3x \frac{e^2}{16\pi^2} \cdot \theta(\mathbf{r}, t) \cdot {}^* F^{\mu\nu} F_{\mu\nu} \quad (119)$$

We integrate with respect to the variable ϕ from 0 to 1. Due to the invariance of the partition function, we arrive at the following expression for S :

$$S = \int dt d^3x \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi + \frac{e^2}{16\pi^2} \int dt d^3x \theta(\mathbf{r}, t) {}^* F^{\mu\nu} F_{\mu\nu} \quad (120)$$

where the first term represents the (trivial) action of massless Dirac fermions, and the second term is the θ term with $\theta(\mathbf{r}, t) = 2(\mathbf{b} \cdot \mathbf{r} - \mu_5 t)$.

3.1 Anomalous Hall effect and negative magnetoresistance

Let us consider the consequences of the presence of a θ term in Weyl semimetals. As we also saw for insulators with a dynamical axion field, an electric current is induced in the presence of a θ term. The induced current density and charge density are given by:

$$\begin{cases} \mathbf{j}(\mathbf{r}, t) = -\frac{\alpha}{\pi} (\dot{\theta} \cdot \mathbf{B} + \nabla \theta \times \mathbf{E}) \\ \rho(\mathbf{r}, t) = \frac{\alpha}{\pi} (\nabla \theta) \cdot \mathbf{B} \end{cases} \quad (121)$$

In the present case $\theta(\mathbf{r}, t) = 2(\mathbf{b} \cdot \mathbf{r} - \mu_5 t)$, we readily obtain a static current of the form:

$$\mathbf{j} = \frac{2\alpha}{\pi}(\mu_5 \mathbf{B} - \mathbf{b} \times \mathbf{E}) \quad (122)$$

The second term corresponds to the anomalous Hall effect. Consider a 2D plane perpendicular to $\mathbf{b}$. Setting $\mathbf{b} = (0, 0, b)$, we obtain the 3D Hall conductivity:

$$\sigma_{xy}^{3D} = \frac{2\alpha b}{\pi} \quad (123)$$

The first term corresponds to the chiral magnetic effect. It indicates that a direct current is generated along a static magnetic field even in the absence of an electric field, provided there exists a chemical-potential difference $\delta\mu = 2\mu_5$ between the two Weyl nodes. However, an equilibrium (static) chiral magnetic effect is ruled out in crystalline solids. It can nevertheless be realized out of equilibrium, when the system is driven away from equilibrium by the combined effect of electric and magnetic fields. In this case, a chemical-potential difference can be generated by the chiral anomaly. The chiral anomaly in Weyl semimetals refers to the nonconservation of electron number in a given Weyl cone under parallel electric and magnetic fields; the rate of pumping of electrons is given by:

$$\frac{\partial N_i}{\partial t} = Q_i \frac{\alpha}{\pi} \mathbf{E} \cdot \mathbf{B} \quad (124)$$

where i is a valley (Weyl node) index, and

$$Q_i = \int \frac{d^3k}{2\pi} \cdot \frac{\partial f_0(\mathcal{E}_k^m)}{\partial \mathcal{E}_k^m} \cdot \mathbf{v}_k^m \cdot \mathbf{\Omega}_k^m \quad (125)$$

is the chirality of the valley. Here, $\mathcal{E}_k^m$ is the energy of Bloch electrons with momentum $\mathbf{k}$ in band m in valley i , and $f_0(\mathcal{E}_k^m)$ is the Fermi distribution function. $\mathbf{v}_k^m$ is the group velocity and $\mathbf{\Omega}_k^m$ is the Berry curvature. Therefore, a difference in the total electron number between the Weyl nodes leads to a difference in the chemical potential between the nodes.

One phenomenon associated with the chiral anomaly is negative magnetoresistance:

$$\sigma_{zz} = \frac{\alpha}{\pi} \frac{(eB_z)^2 v_F^3}{\mu^2} \tau_{\text{inter}} \quad (126)$$

where μ is the equilibrium chemical potential and τ_{inter} is the intervalley scattering time. This unusual magnetoconductivity holds in the low-field limit $B_z \rightarrow 0$. The expression for $\mathbf{j}$ can be understood as coming from:

$$\mathbf{j} \propto (\mathbf{E} \cdot \mathbf{B} \tau_{\text{inter}}) \cdot \mathbf{B} \quad (127)$$

which indicates that it is a consequence of the chiral magnetic effect.